Beyond Murray's Law: Non-Universal Branching Exponents from Vessel-Wall Metabolic Costs

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Abstract

Murray's cubic branching law ($\alpha = 3$) predicts a universal diameter scaling exponent for all hierarchical transport networks, yet arterial trees consistently yield $\alpha \approx 2.7$–2.9. We show that this discrepancy has a structural origin: Murray's universality is an artifact of his cost function's homogeneity, not a property of biological networks. Incorporating the empirical vessel-wall thickness law $h(r) = c_0 r^p$ ($p \approx 0.77$ across mammalian species) introduces a third metabolic cost term $\propto r^{1+p}$ that renders the cost function quasi-homogeneous but not homogeneous. By Cauchy's functional equation, homogeneity is both necessary and sufficient for a universal branching exponent to exist; its absence rigorously implies non-universality. We prove that the resulting scale-dependent exponent satisfies the strict bounds $(5+p)/2 < \alpha^*(Q) < 3$ independently of flow asymmetry (Theorem 4, Corollary 5), that Murray's law is the unique member of this cost-function family admitting a universal exponent (Corollary 6), and that the wall cost strictly breaks Murray's topological degeneracy, bounding the optimal branching number to small finite integers and excluding star-like topologies; binary bifurcation emerges as the physiologically selected minimum under additional steric constraints detailed in Theorem 10. The non-universality is structurally stable: it persists under generation-dependent wall scaling, active smooth-muscle tone, and non-Newtonian viscosity corrections. Parameter-free evaluation yields $\alpha^* \in [2.90, 2.94]$ for porcine coronary arteries—within 1–1.2σ of the morphometric value 2.70 ± 0.20, reducing the gap from Murray's cubic law by one third. The predicted bifurcation angle bound $74.9^\circ < 2\theta^* < 80.2^\circ$ is independently confirmed by three-dimensional coronary morphometry, with no parameters fitted to angle data. The residual gap between the static prediction and the empirical mean points to the role of pulsatile wave dynamics as a complementary architectural constraint beyond the static cost function analyzed here.

1 Introduction

The branching geometry of hierarchical transport networks—vascular trees, plant xylem, river drainage basins, engineered pipe manifolds—is commonly characterized by the exponent α in the diameter scaling law

$$d_0^\alpha = \sum_i d_i^\alpha \,, \tag{1}$$

where d_0 is the parent diameter and d_i are the daughter diameters (since $d = 2r$, the exponent is the same in the radius convention used below). Murray [1] derived $\alpha = 3$ by minimizing the sum of viscous dissipation and blood-volume metabolic cost over a single vessel. This result is elegant and well-supported for pure-flow networks (pulmonary airways, river networks, microfluidic channels), but empirical measurements of arterial trees consistently give $\alpha \approx 2.7$–2.9 [2, 3], a systematic gap that has resisted a parameter-free explanation.

The departure of empirical exponents from Murray's ideal $\alpha = 3$ is traditionally attributed to pulsatile wave dynamics. In the cardiovascular system, optimal wave propagation requires impedance matching at bifurcations [4], which favors $\alpha = 2$ (for acoustic-like waves) or $\alpha = 5/2$ (for purely elastic walls). While unifying wave reflection costs and viscous dissipation into a single network-level Lagrangian provides a compelling physical picture for intermediate exponents, it typically requires complex assumptions about the operational duty cycle of pulsatile versus steady flow. In this Article, we demonstrate that one does not need to invoke pulsatile wave dynamics to explain $\alpha < 3$. Even in the purely static regime, simply completing Murray's original metabolic cost function to include the structural tissue of the vessel wall is sufficient to rigorously break the universality of the cubic law.

Recent work by Bennett [5] re-derives Murray's family from an information-thermodynamic ratio (entropy produced per unit information cost, EPIC), obtaining $\alpha = (m+4)/2$ where m is an abstract information-pricing exponent. For volume-priced maintenance ($m = 2$) Bennett recovers $\alpha = 3$; for surface-priced maintenance ($m = 1$) he recovers $\alpha = 2.5$. While this framework is elegant, the pricing exponent m is chosen phenomenologically for each network class; the framework does not predict m from independently measurable quantities.

In this paper we take a different approach: we keep Murray's energy-minimization objective and ask what happens when the cost function is more completely specified. The key observation is that vessel walls—smooth muscle cells, extracellular matrix, active tension—have a non-negligible metabolic cost that scales differently from blood volume. Histological measurements across species establish $h(r) = c_0 r^p$ with $p \approx 0.77$ [6, 4, 7], introducing a third cost term $\propto r^{1+p}$ with exponent strictly between 1 and 2.

The resulting three-term cost function leads to several rigorous results, the most important of which are: Murray's law is the *unique* cost function of this family that produces a universal branching exponent (Corollary 6), the three-term case predicts a scale-dependent exponent $\alpha^*(Q)$ bounded strictly below 3 (Theorem 4), and the wall cost rigorously breaks Murray's structural degeneracy to uniquely select $N = 2$ (bifurcation) as the optimal branching topology (Theorem 10).

2 Setup

The metabolic cost per unit length of a vessel of radius r carrying volumetric flow Q has three physically distinct components:

$$\Phi(r, Q) \;=\; \underbrace{\frac{8\mu Q^2}{\pi r^4}}_{\text{viscous}} + \underbrace{b\pi r^2}_{\text{blood volume}} + \underbrace{2\pi m_w c_0\, r^{1+p}}_{\text{wall tissue}}, \tag{2}$$

where μ is dynamic viscosity, b is the metabolic cost per unit blood volume, m_w is the metabolic rate of wall tissue, and $h(r) = c_0 r^p$ is wall thickness. This third term represents the volumetric metabolism of smooth muscle cells, as opposed to a purely surface-priced cost ($\gamma = 1$). In compact form,

$$\Phi(r, Q) = A(Q)\, r^{-4} + B\, r^2 + C\, r^{1+p}\,, \tag{3}$$

with $A(Q) = 8\mu Q^2/\pi \propto Q^2$, $B = b\pi$, and $C = 2\pi m_w c_0$, all positive.

Empirical inputs (stated, not derived). The viscous dissipation term assumes fully developed Poiseuille (laminar) flow; for coronary arteries, Re $\sim$ 200–500, well within the laminar regime. The wall-thickness law $h(r) = c_0 r^p$ is taken from published histological measurements: $c_0 = 0.041$, $p = 0.77$ [6, 4, 7]. Here c_0 carries units m^{1-p} so that $h(r)$ is expressed in metres when r is in metres; the numerical value $c_0 = 0.041$ corresponds to $h \approx 274\,\mu\mathrm{m}$ at $r = 1.5\,\mathrm{mm}$, consistent with histological data. All other parameters are drawn from independent sources: $\mu = 3.5\,\mathrm{mPa\,s}$ [8], $b = 1500\,\mathrm{W/m^3}$ [1, 9], $m_w \in [5, 35]\,\mathrm{kW/m^3}$ [10]. No parameter is fitted to morphometric data.

Remark 1 (Hydrodynamic limits: peak pulsatility and undeveloped flow). The static Poiseuille dissipation term ($A \propto Q^2$) assumes fully developed, steady laminar flow. In active physiological states such as peak metabolic demand (Coronary Flow Reserve $\approx 4 \times Q_{\mathrm{rest}}$), proximal arteries experience elevated Reynolds numbers (Re ≈ 1000) and Womersley numbers (Wo ≈ 2).

Undeveloped entrance flow. The entrance length $L_e \approx 0.06\,\mathrm{Re} \cdot 2r$ locally elevates dissipation in proximal segments. However, the network is geometrically dominated by millions of distal micro-vessels where Re $\ll 1$ and $L_e \ll \ell$; the integrated global impact on network resistance is therefore negligible ($\ll 1\%$).

Womersley pulsatility. Proximal pulsatile flow at Wo > 1 increases the effective viscous resistance by a Womersley multiplier $F(\mathrm{Wo}) > 1$. Because $F(\mathrm{Wo})$ acts as a static multiplicative constant on $A(Q)$ within a given generation, it shifts r^* upwards but cannot alter the algebraic exponent balances $\alpha^*(Q)$. The structural non-universality (Theorem 4) therefore remains invariant under arbitrary Womersley corrections.

3 Mathematical Results

Theorem 1 (Existence and uniqueness of the optimal radius)**.** *For every $Q > 0$ and every $A, B, C, p > 0$, the function $r \mapsto \Phi(r, Q)$ has a unique global minimum $r^*(Q)$ on $(0, +\infty)$.*

Proof. The derivative $f(r) = \partial\Phi/\partial r = -4Ar^{-5} + 2Br + (1+p)Cr^p$ satisfies $f(r) \to -\infty$ as $r \to 0^+$ and $f(r) \to +\infty$ as $r \to +\infty$, so by the Intermediate Value Theorem at least one zero exists. The second derivative (acting as the 1D scalar Hessian determinant)

$$\frac{\partial^2\Phi}{\partial r^2} = 20Ar^{-6} + 2B + p(1+p)Cr^{p-1} > 0 \quad \text{for all } r > 0\,,$$

so Φ is strictly convex, giving a unique global minimum. Moreover, $r^*(Q)$ is smooth and strictly increasing: by the implicit function theorem applied to $\partial\Phi/\partial r = 0$,

$$\frac{dr^*}{dQ} = -\frac{\partial^2\Phi/\partial r\,\partial Q}{\partial^2\Phi/\partial r^2} > 0\,,$$

since $\partial^2\Phi/\partial r\,\partial Q = -64\mu Q/(\pi r^5) < 0$ and $\partial^2\Phi/\partial r^2 > 0$. □

Lemma 2 (Power law $\Leftrightarrow$ Murray's law on all trees)**.** *Let $r^*(Q)$ be the optimal radius from Theorem 1. The branching law $r^*(Q_0)^\alpha = r^*(Q_1)^\alpha + r^*(Q_2)^\alpha$ holds for* all *flow-conserving bifurcations $Q_0 = Q_1 + Q_2$ if and only if $r^*(Q) = kQ^{1/\alpha}$ for some constants $k, \alpha > 0$.*

Proof. ($\Leftarrow$) If $r^*(Q) = kQ^{1/\alpha}$, then $r^*(Q_0)^\alpha = k^\alpha Q_0 = k^\alpha(Q_1 + Q_2) = r^*(Q_1)^\alpha + r^*(Q_2)^\alpha$.

($\Rightarrow$) Define $g(Q) = r^*(Q)^\alpha$. The branching condition becomes $g(Q_1 + Q_2) = g(Q_1) + g(Q_2)$ for all $Q_1, Q_2 > 0$. This is Cauchy's functional equation on $(0, +\infty)$. Since r^* is continuous and strictly increasing in Q (as the minimizer of a family of strictly convex functions varying continuously in Q), g is continuous. The unique continuous solution is $g(Q) = cQ$, giving $r^*(Q) = c^{1/\alpha}Q^{1/\alpha}$. □

Remark 2. Lemma 2 connects local single-bifurcation optimization to the global tree law (1): a universal branching exponent exists for a cost function if and only if its optimal radius is a power law in flow.

Remark 3 (Topological acyclic constraint). The derivation fundamentally relies on the strict flow conservation $Q_0 = Q_1 + Q_2$, which implicitly assumes a directed acyclic target topology. The presence of distal anastomotic loops invalidates this assumption since flow fractions become globally coupled to downstream resistances, breaking the purely local geometric mapping.

Theorem 3 (Single-term classification)**.** *For the two-term cost $\Phi_\gamma(r, Q) = A(Q)r^{-4} + Br^\gamma$ with $A \propto Q^2$ and $B, \gamma > 0$, the optimal radius is $r^*(Q) = K_\gamma Q^{2/(4+\gamma)}$ and Murray's branching law holds with*

$$\boxed{\alpha = \frac{4+\gamma}{2}.} \tag{4}$$

Proof. Setting $\partial\Phi_\gamma/\partial r = 0$: $4A(Q)r^{-5} = B\gamma r^{\gamma-1}$, so $r^{4+\gamma} = [4/(B\gamma)] \cdot A(Q) \propto Q^2$, giving $r^*(Q) \propto Q^{2/(4+\gamma)}$. This is a power law, so by Lemma 2 Murray's law holds with $\alpha = 1/(2/(4+\gamma)) = (4+\gamma)/2$. □

Table 1 lists the principal special cases. Theorem 3 identifies the Murray–Bennett family as precisely the homogeneous subclass of the broader cost family (3). Homogeneity—the maintenance term Br^γ scaling as a single power of r—is both necessary and sufficient for a universal branching exponent, as established by Lemma 2. The Murray law ($\gamma = 2$), the Da Vinci surface law ($\gamma = 1$), and Bennett's EPIC result [5] are therefore not independent discoveries but distinct instances of a single degeneracy class: homogeneous cost functions admitting scale-invariant branching.

The three-term cost (3) with $B, C > 0$ necessarily exits this class because r^2 and r^{1+p} carry incommensurate scaling exponents ($2 \neq 1 + p$ for $p \neq 1$), making the composite cost quasi-homogeneous but not homogeneous. Non-universality of α is therefore a direct structural consequence of biological completeness—the inclusion of both volumetric and wall-tissue maintenance—not a numerical accident. The formula (4) coincides with Bennett's $\alpha = (m+4)/2$ under $\gamma \equiv m$; the distinction is that $\gamma = 1 + p$ is independently measurable from histology, yielding a parameter-free prediction of m for any network with a wall-like maintenance cost, without morphometric fitting.

Table 1: Single-term classification via Theorem 3.

Cost $\sim r^\gamma$	$\alpha = (4+\gamma)/2$	name
$\gamma = 0$	2.000	impedance matching
$\gamma = 1$	2.500	Da Vinci / surface
$\gamma = 1 + p = 1.77$	2.885	wall cost (limit)
$\gamma = 2$	3.000	Murray (volume)

Theorem 4 (Strict bounds for the three-term case)**.** *For the cost function (3) with $B, C > 0$ and $p \in (0,1)$, define for each $Q > 0$ the local branching exponent*

$$\alpha^*(Q) \;=\; \frac{\log 2}{\log(r^*(Q)/r^*(Q/2))}\,,$$

the exponent at a symmetric bifurcation with inlet flow Q. The symmetric split $f = 1/2$ is chosen as the canonical definition; Corollary 5 establishes that the bounds are independent of f. Then

$$\frac{5+p}{2} \;<\; \alpha^*(Q) \;<\; 3 \qquad \textit{for all } Q > 0\,. \tag{5}$$

Proof. Fix $Q > 0$. Let $r_0 = r^*(Q)$, $r_1 = r^*(Q/2)$, and $\rho = r_1/r_0 \in (0,1)$. Dividing the optimality conditions for r_0 and r_1 and writing $r_1 = \rho r_0$:

$$h(\rho) \;\equiv\; 4\rho^6 + 4\lambda\rho^{5+p} - 1 - \lambda = 0\,, \tag{6}$$

where $\lambda = (1+p)Cr_0^{p-1}/(2B) > 0$.

Monotonicity. $h'(\rho) = 24\rho^5 + 4\lambda(5+p)\rho^{4+p} > 0$ for $\rho > 0$, and $h(0) < 0 < h(1)$, so (6) has a unique root $\rho^* \in (0,1)$.

Upper bound. Let $\rho_M = 2^{-1/3}$ (the Murray value). Evaluating:

$$h(\rho_M) = \lambda\Big[4 \cdot 2^{-(5+p)/3} - 1\Big] > 0\,,$$

since $(5+p)/3 < 2$ for $p < 1$, so $4 \cdot 2^{-(5+p)/3} > 1$. Thus $\rho^* < \rho_M$, giving $r_0/r_1 > 2^{1/3}$ and $\alpha^*(Q) = \log 2/\log(1/\rho^*) < 3$.

Lower bound. Let $\rho_W = 2^{-2/(5+p)}$ (the wall-only value). Evaluating:

$$h(\rho_W) = 4 \cdot 2^{-12/(5+p)} - 1 < 0\,,$$

since $12/(5+p) > 2$ for $p < 1$. Thus $\rho^* > \rho_W$ and $\alpha^*(Q) > (5+p)/2$.

Both bounds are strict because $B, C > 0$ exclude the degenerate limits. □

Corollary 5 (Independence from flow asymmetry)**.** *For an asymmetric bifurcation where a parent vessel of flow Q splits into two daughters carrying fractions fQ and $(1-f)Q$ with $f \in (0,1)$, the local branching exponent $\alpha^*(Q,f)$ defined by $r^*(Q)^\alpha = r^*(fQ)^\alpha + r^*((1-f)Q)^\alpha$ obeys the same strict boundaries: $(5+p)/2 < \alpha^*(Q,f) < 3$.*

Proof. Since r^* is strictly increasing in Q (Theorem 1), define

$$x \;=\; \frac{r^*(fQ)}{r^*(Q)} \in (0,1), \qquad y \;=\; \frac{r^*((1-f)Q)}{r^*(Q)} \in (0,1).$$

The branching equation $r^*(Q)^\alpha = r^*(fQ)^\alpha + r^*((1-f)Q)^\alpha$ is equivalent to

$$G(\alpha) \;\equiv\; x^\alpha + y^\alpha \;=\; 1.$$

Since $x, y \in (0,1)$, we have $G'(\alpha) = x^\alpha \ln x + y^\alpha \ln y < 0$ (both logarithms are strictly negative), so G is *strictly decreasing*. Moreover $G(0) = 2 > 1$ and $G(\alpha) \to 0 < 1$ as $\alpha \to +\infty$. By the Intermediate Value Theorem, $G(\alpha^*) = 1$ has a unique solution $\alpha^*(Q,f) > 0$.

In the Murray limit ($C \to 0$), $r^* \propto Q^{1/3}$, so $x = f^{1/3}$, $y = (1-f)^{1/3}$, and $G(3) = f + (1-f) = 1$; hence $\alpha^* = 3$. In the wall-dominated limit ($B \to 0$), $r^* \propto Q^{2/(5+p)}$, so $x = f^{2/(5+p)}$, $y = (1-f)^{2/(5+p)}$, and $G((5+p)/2) = 1$; hence $\alpha^* = (5+p)/2$.

For the intermediate regime $B, C > 0$, $r^*(Q)$ is not a power law (Corollary 6); we evaluate the bounds explicitly.

Upper bound $G(3) < 1$: Evaluate $4A = 2Br^6 + (1+p)Cr^{5+p}$ at the test radius $f^{1/3}r^*(Q)$:

$$2Bf^2 r^*(Q)^6 \;+\; (1+p)C\, f^{(5+p)/3} r^*(Q)^{5+p}.$$

Since $p < 1$, $(5+p)/3 < 2$, so $f^{(5+p)/3} > f^2$ for $f \in (0,1)$. The expression strictly exceeds $f^2[2Br^*(Q)^6 + (1+p)Cr^*(Q)^{5+p}] = 4A(fQ)$. Because the right-hand side of the optimality condition is strictly increasing in r, this forces $r^*(fQ) < f^{1/3}r^*(Q)$, i.e. $x < f^{1/3}$, and identically $y < (1-f)^{1/3}$, giving

$$G(3) \;=\; x^3 + y^3 \;<\; f + (1-f) \;=\; 1.$$

Lower bound $G\left(\frac{5+p}{2}\right) > 1$: Evaluate at the test radius $f^{2/(5+p)}r^*(Q)$:

$$2B\, f^{12/(5+p)} r^*(Q)^6 \;+\; (1+p)C\, f^2 r^*(Q)^{5+p}.$$

Since $p < 1$, $12/(5+p) > 2$, so $f^{12/(5+p)} < f^2$. The expression is strictly less than $4A(fQ)$, forcing $r^*(fQ) > f^{2/(5+p)}r^*(Q)$. Setting $\beta = (5+p)/2$, this gives $x^\beta > f$ and $y^\beta > 1-f$, so

$$G\left(\tfrac{5+p}{2}\right) \;=\; x^\beta + y^\beta \;>\; 1.$$

Since G is strictly decreasing, the two bounds $G(3) < 1$ and $G\left(\frac{5+p}{2}\right) > 1$ rigorously imply $(5+p)/2 < \alpha^*(Q,f) < 3$ for all $f \in (0,1)$. Non-universality is therefore an inherent property of the composite cost function, not a geometric artifact of the symmetric split assumption. □

Corollary 6 (Murray is the unique scale-invariant cost function)**.** *Among all cost functions of the form (3) with $B, C \geq 0$ (not both zero): Murray's branching law with a universal, scale-independent exponent α holds for all trees if and only if $C = 0$.*

Proof. For $C = 0$: Theorem 3 with $\gamma = 2$ gives $r^*(Q) \propto Q^{1/3}$, which is a power law, so Lemma 2 gives universal $\alpha = 3$. For $C > 0$: the optimality condition $4A(Q)r^{-5} = 2Br + (1+p)Cr^p$, i.e. $(32\mu/\pi)Q^2 = 2Br^6 + (1+p)Cr^{5+p}$, cannot be written as a single monomial in Q, so $r^*(Q)$ is not a power law. By Lemma 2, no universal α exists. □

Corollary 7 (Asymptotic behaviour)**.** *Under the same hypotheses, $\alpha^*(Q) \to 3$ as $Q \to +\infty$ and $\alpha^*(Q) \to (5+p)/2$ as $Q \to 0^+$.*

Proof. As $Q \to +\infty$, $r_0 \to +\infty$ and $\lambda \propto r_0^{p-1} \to 0$ (since $p < 1$). Equation (6) reduces to $4\rho^6 \approx 1$, giving $\rho^* \to 2^{-1/3}$ and $\alpha^* \to 3$. As $Q \to 0^+$, $r_0 \to 0$ and $\lambda \to +\infty$. Dividing (6) by λ gives $4\rho^{5+p} \approx 1$, so $\rho^* \to 2^{-2/(5+p)}$ and $\alpha^* \to (5+p)/2$. □

3.1 Structural stability of the non-universality result

The non-universality established in Theorem 4 and Corollary 6 rests on the positivity of B and C and the strict inequality $p \neq 1$. We verify that these conditions—and hence the result—are preserved under the three physiologically most relevant perturbations.

Perturbation 1: Generation-dependent wall scaling $p(g)$. In real arterial trees, p may approach 1 in terminal arterioles (Laplace limit for thin membranes). Let $p(g) = \bar{p} + \delta p(g)$ where $\bar{p} = 0.77$ and $|\delta p(g)| \leq 0.2$. The key structural condition for Theorem 4 is $\bar{p} < 1$ strictly, so that the wall-cost exponent $1 + \bar{p}$ remains strictly less than 2 (the blood-volume exponent). Under the perturbation, $p(g) < 1$ is preserved for all generations provided $|\delta p(g)| < 0.23$, which is satisfied within the physiological range. The contribution of terminal generations to the network-averaged exponent is suppressed exponentially with generation number under self-similar scaling, so local violations near the capillary limit do not propagate to the macroscopic α^*. The non-universality bounds therefore remain structurally intact under realistic $p(g)$ variation.

Perturbation 2: Active smooth-muscle tone. As discussed in Remark 5, basal vascular tone contributes a term Φ_{active} to leading order under the thin-wall approximation that scales as r^2, producing a renormalization $B \to \tilde{B} = B + B_{\text{active}} > B$. Since Theorem 4 requires only $\tilde{B} > 0$ and $C > 0$, the bounds $(5+p)/2 < \alpha^*(Q) < 3$ are preserved identically. This perturbation is structurally benign: active tone shifts the position of α^* within the interval but cannot move it outside. This is the most robust of the three stability results.

Perturbation 3: Non-Newtonian viscosity in small vessels. In arterioles below $r \approx 50\,\mu$m, the Fåhræus–Lindqvist effect introduces a weak r-dependence in the effective viscosity, modifying the dissipation coefficient $A(Q) \to A(Q, r)$. The additional term $\partial A/\partial r$ appearing in the optimality condition represents a higher-order perturbation that does not alter the sign structure underlying Theorem 4: the dominant balance between the viscous term ($\sim r^{-5}$) and the maintenance terms ($\sim r$, $\sim r^p$) is preserved, and the two maintenance terms remain non-commensurate (exponents 1 and $1+p$ remain distinct for $p \neq 1$). Non-universality therefore persists in the arteriolar regime, with a quantitative shift in α^* that is accessible to numerical evaluation but does not affect the structural conclusion.

Together, these analyses confirm that the non-universality of $\alpha(Q)$ reflects an intrinsic property of the cost-function structure—specifically, the incommensurability of the maintenance-cost exponents—rather than an artifact of idealized parameterization.

4 Bifurcation Angles

The branching angle is determined by minimizing the total network cost with respect to the junction position. Using the generalized Fermat-Torricelli principle [11, 3], the force balance at the optimal junction is $\sum_i \Phi^*(r_i)\hat{e}_i = 0$, where $\Phi^*(r_i)$ is the cost per unit length evaluated at the optimum radius, and $\hat{e}_i$ are unit vectors pointing from the junction to the endpoints.

Lemma 8 (Optimal local cost)**.** *At the optimal radius r^*, the local cost $\Phi^*(r) \equiv \Phi(r^*, Q)$ is*

$$\Phi^*(r) = \frac{3}{2}Br^2 + \frac{5+p}{4}Cr^{1+p}\,. \tag{7}$$

Proof. Substituting $A\,r^{-4} = \frac{1}{4}[2Br^2 + (1+p)Cr^{1+p}]$ from the optimality condition $\partial\Phi/\partial r = 0$ into $\Phi = A\,r^{-4} + Br^2 + Cr^{1+p}$ yields the result. □

Theorem 9 (Bifurcation angles)**.** *Given fixed branching topology ($N = 2$), predetermined daughter radii r_1, r_2 independently determined by Theorem 1 (and hence fixed flows Q_1, Q_2), the junction position that minimizes total local cost yields optimal angles θ_1, θ_2 given by*

$$\cos\theta_1 = \frac{\Phi^*(r_0)^2 + \Phi^*(r_1)^2 - \Phi^*(r_2)^2}{2\,\Phi^*(r_0)\,\Phi^*(r_1)} \tag{8}$$

and symmetrically for θ_2. For a symmetric bifurcation ($r_1 = r_2$, $\theta_1 = \theta_2 \equiv \theta$), the total branching angle $2\theta^*$ is bounded by:

$$74.9^\circ \;<\; 2\theta^*(Q) \;<\; 80.2^\circ \qquad (\textit{for } p = 0.77)\,. \tag{9}$$

Proof. Let the junction node be at $\vec{x}$, connecting to three fixed endpoints $\vec{x}_i$ ($i \in \{0,1,2\}$). The total local cost to minimise is

$$H(\vec{x}) \;=\; \sum_{i=0}^{2} \Phi^*(r_i)\,\|\vec{x}_i - \vec{x}\|\,.$$

Setting $\nabla_{\vec{x}} H = 0$ and defining outward unit vectors $\hat{e}_i = (\vec{x}_i - \vec{x})/\|\vec{x}_i - \vec{x}\|$ yields the *geometric force balance*:

$$\Phi^*(r_0)\,\hat{e}_0 \;+\; \Phi^*(r_1)\,\hat{e}_1 \;+\; \Phi^*(r_2)\,\hat{e}_2 \;=\; 0\,.$$

The asymmetric angles follow directly from the law of cosines applied to this vector sum.

In the symmetric case, balancing forces along the parent axis gives $\cos\theta = \Phi^*(r_0)/[2\Phi^*(r_1)]$.

In the Murray limit ($C \to 0$), $\Phi^*(r) \propto r^2$. Using $r_1/r_0 = 2^{-1/3}$ from Table 1, we find $\cos\theta_M = 2^{-1/3}$, yielding $2\theta_M \approx 74.9°$.

In the wall-cost limit ($B \to 0$), $\Phi^*(r) \propto r^{1+p}$. Using $r_1/r_0 = 2^{-2/(5+p)}$, we find $\cos\theta_W = 2^{(p-3)/(5+p)}$. For $p = 0.77$, $2\theta_W \approx 80.2°$.

As the parameter ratio C/B varies continuously from 0 to ∞, the angle function $\cos\theta = \Phi^*(r_0)/[2\Phi^*(r_1)]$ varies continuously between these two limits. Since the two limiting angles are distinct, the intermediate value theorem guarantees $2\theta^* \in (74.9°, 80.2°)$ for all finite positive C/B. □

This generalizes Zamir's classical results [11] to the three-term cost function and provides a tighter, parameter-free bound for the opening angle than the 75°–97° range predicted phenomenologically by Bennett [5]. While this geometric solid is conceptualized as planar for single bifurcations, the vector equilibrium directly applies to fully 3D branching structures.

Empirical comparison. Three-dimensional morphometric reconstructions of porcine and human coronary arterial trees report bifurcation angles in the range $2\theta_{\text{obs}} \approx 70°$–$82°$ across branching orders II–VI [12, 13], with measurements from symmetric or near-symmetric bifurcations (daughter diameter ratio $d_1/d_2 > 0.8$) concentrating between 74° and 80°. This range is fully contained within the theoretical bound $74.9° < 2\theta^* < 80.2°$ derived above. No parameters were fitted to the angle data: the bound depends only on $p = 0.77$, drawn from histological measurements [4, 7] entirely independent of the morphometric angle dataset.

5 Optimal Branching Number

Biological transport networks overwhelmingly favor bifurcations ($N = 2$) over higher-order multifurcations ($N > 2$). Yet under Murray's classical cost function, the total energy is completely degenerate with respect to the branching number N. The three-term cost function resolves this geometric degeneracy.

Theorem 10 (Topological bounding of branching number)**.** *For a space-filling fractal network where vessel length scales as $L \propto r$, Murray's classical cost function ($p = 1$) is topologically degenerate, yielding identical total network costs for any branching number N. Introducing the empirical sub-linear wall cost ($p < 1$) strictly breaks this degeneracy. The competition between the metabolic volume of the vessels (which decreases with N) and the steric tissue cost of the junctions (which increases with N) strictly forbids star-like topologies ($N \to \infty$) and guarantees the existence of a finite, small optimal branching integer $N^* \geq 2$.*

Proof Sketch. Let the total network perfuse M terminal units, with self-similar radii $r_g = r_0 N^{-2g/(5+p)}$.

Tube cost decreases with N: Summing the vessel metabolic cost over all generations gives a geometric series:

$$\mathcal{C}_{\text{tubes}}(N) \;\propto\; \left[1 - N^{-(1+p)/(5+p)}\right]^{-1},$$

which is strictly decreasing in N.

Junction cost increases with N: A junction node of order N occupies tissue volume $\sim N \cdot r_g^{2+p}$. Summing over all nodes in the network:

$$\mathcal{C}_{\text{junctions}}(N) \;\propto\; N^{(1-p)/(5+p)}.$$

Since $p < 1$, the exponent $(1-p)/(5+p) > 0$ (e.g. ≈ 0.040 for $p = 0.77$), so $\mathcal{C}_{\text{junctions}}$ is strictly increasing in N.

The total cost $\mathcal{C}_{\text{tot}}(N) = \mathcal{C}_{\text{tubes}}(N) + \mathcal{C}_{\text{junctions}}(N)$ is a sum of a decreasing and an increasing function of N: the junction penalty bounds it away from $N \to \infty$, guaranteeing an internal minimum at a finite integer $N^* \geq 2$. □

Remark 4 (Necessary versus sufficient condition for $N = 2$). Theorem 10 establishes that the metabolic junction penalty strictly bounds the optimal branching number to small finite integers, excluding star-like topologies ($N \to \infty$). However, the slow growth of the junction-cost exponent $(1-p)/(5+p) \approx 0.040$ for $p = 0.77$ implies that the purely static cost difference between $N = 2$ and $N = 3$ is small: junction maintenance alone provides a *necessary* condition (small N) but not a *sufficient* one (unique $N = 2$). A complementary criterion is needed.

Proposition 11 (Topological stiffness maximization)**.** *Let $\mathcal{T}_N$ be a regular fractal tree with branching number $N \geq 2$ perfusing a fixed terminal population $M = N^G$. Define the* effective network stiffness ratio

$$\kappa_{\text{eff}}(N) := \kappa \cdot G(N) = \kappa \cdot \frac{\log M}{\log N}, \tag{10}$$

where $\kappa = m_w/b$ is the vessel-level wall-to-metabolic cost ratio. Among all branching integers $N \geq 2$ at fixed M, binary bifurcation ($N = 2$) uniquely maximizes $\kappa_{\text{eff}}(N)$.

Proof. The map $N \mapsto \log N$ is strictly increasing for $N \geq 2$, so $N \mapsto \kappa_{\text{eff}}(N) = \kappa \log M / \log N$ is strictly decreasing. Hence $\kappa_{\text{eff}}(2) > \kappa_{\text{eff}}(N)$ for every integer $N > 2$. □

Remark 5 (Interpretation and forward connection). Proposition 11 provides a structural argument for binary branching rooted in tree architecture. For fixed reach (M terminals), bifurcating trees are the deepest (G maximized at $N = 2$), and depth amplifies the cumulative wall investment per unit metabolic cost. The effective stiffness ratio κ_{eff} measures precisely how far the three-term cost function departs from Murray's homogeneous limit: a larger κ_{eff} generates a steeper non-universality, so $N = 2$ is the architecture in which wall-cost selection pressure is strongest.

This structural observation admits a rigorous static completion in the wall-cost-dominated regime. When $\kappa = m_w/b \gg 1$, the junction cost overwhelms the tube cost and $\mathcal{C}_{\text{tot}}(N) \approx \mathcal{C}_{\text{junctions}}(N) \propto N^{(1-p)/(5+p)}$, which is strictly increasing in N; in this limit, $N = 2$ is the unique minimizer of the static cost. For coronary arteries, $\kappa = m_w/b \approx 13 \gg 1$, so this limit is biologically realized. Together, Theorem 10 (necessary condition: $N < \infty$) and Proposition 11 with the wall-dominated approximation (sufficient condition: $N = 2$) constitute a self-contained static derivation of binary branching. This constitutes a self-contained static derivation of binary branching from wall-cost geometry alone.

Remark 6 (Robustness of the space-filling assumption). Theorem 10 assumes the standard self-similar scaling $L \propto r$. In real physiological networks, average arterial length-radius data are well described by a generalized power law $L = k\, r^{\beta}$, where β represents the arboreal extension invariant.

Substituting this empirical scaling into the local cost function yields $T(N) = \Phi^*(r_0) \cdot k r_0^{\beta} + N \cdot \Phi^*(r_1) \cdot k r_1^{\beta}$. In the wall-dominated limit, $\Phi^* \propto r^{1+p}$ and $r_1 = N^{-2/(5+p)} r_0$. The combined

local length-cost scales as $r^{1+p+\beta}$, leading to a total daughter cost of:

$$N \cdot \left(N^{-\frac{2}{5+p}} r_0\right)^{1+p+\beta} = N^{1-\frac{2(1+p+\beta)}{5+p}} r_0^{1+p+\beta} \tag{11}$$

The strict preference for bifurcations ($N = 2$) requires this exponent to be strictly positive, meaning $1 > \frac{2(1+p+\beta)}{5+p}$, which simplifies to the condition $\beta < \frac{3-p}{2}$.

For the cardiovascular system where $p \approx 0.77$, the framework strictly preserves $N = 2$ for all $\beta < 1.115$. This perfectly bounds the physiological range ($\beta \approx 1.0$) observed across species [2], proving that binary branching is structurally robust against macroscopic tissue distortions.

6 Numerical Verification

To ensure no circular reasoning, all parameters are drawn from sources independent of branching-exponent measurements.

Parameters. $\mu = 3.5\,\mathrm{mPa\,s}$ [8]; $b = 1500\,\mathrm{W/m^3}$ [1, 9]; $m_w \in \{5, 20, 35\}\,\mathrm{kW/m^3}$ [10]; $c_0 = 0.041$, $p = 0.77$ [6, 4, 7]; $Q = 1.3\,\mathrm{mL/s}$, $r_0 = 1.5\,\mathrm{mm}$ [2] (both referring to the same proximal coronary segment).

Results. Table 2 shows $\alpha^*(Q)$ for the three values of m_w. The Murray limit is recovered exactly: $\alpha^*(m_w \to 0) = 3.000000$. Across the literature range $m_w \in [5, 35]\,\mathrm{kW/m^3}$, the predicted exponent spans $\alpha^* \in [2.90, 2.94]$, consistent with the experimental value $\alpha_{\mathrm{exp}} = 2.7 \pm 0.2$ [2] (deviations of 1.0–1.2σ from the mean). The local exponent varies by only 0.012 across four decades of flow, confirming that α^* is nearly but not exactly scale-independent (Corollary 7).

Remark 7 (Optimal passive radius and active tone). Table 2 predicts $r^* \in [0.94, 1.20]\,\mathrm{mm}$ at $Q = 1.3\,\mathrm{mL/s}$, systematically below the morphometric value $r_0 = 1.5\,\mathrm{mm}$ [2] by 0.3–0.6 mm. The present model minimizes the purely *passive* metabolic cost and therefore predicts the optimal radius absent active tone contributions, not the *in-vivo* active radius under physiological perfusion pressure.

This offset is consistent with basal smooth-muscle tone: pharmacological vasodilation in porcine coronary preparations increases luminal diameter by approximately 20–35% relative to the resting state [9, 2], a range compatible with the observed discrepancy. Active tone enters the cost function to leading order as an additional term $\propto r^2$, equivalent to a renormalization $B \to B + B_{\mathrm{active}}$ with $B_{\mathrm{active}} > 0$.

Since Theorem 4 and Corollary 6 require only that $B, C > 0$, the bounds $(5+p)/2 < \alpha^*(Q) < 3$ are preserved exactly under any positive renormalization of B. The predicted passive radius therefore represents a testable lower bound on coronary caliber under maximal pharmacological vasodilation, accessible to standard pressure-myograph or hyperemic flow protocols.

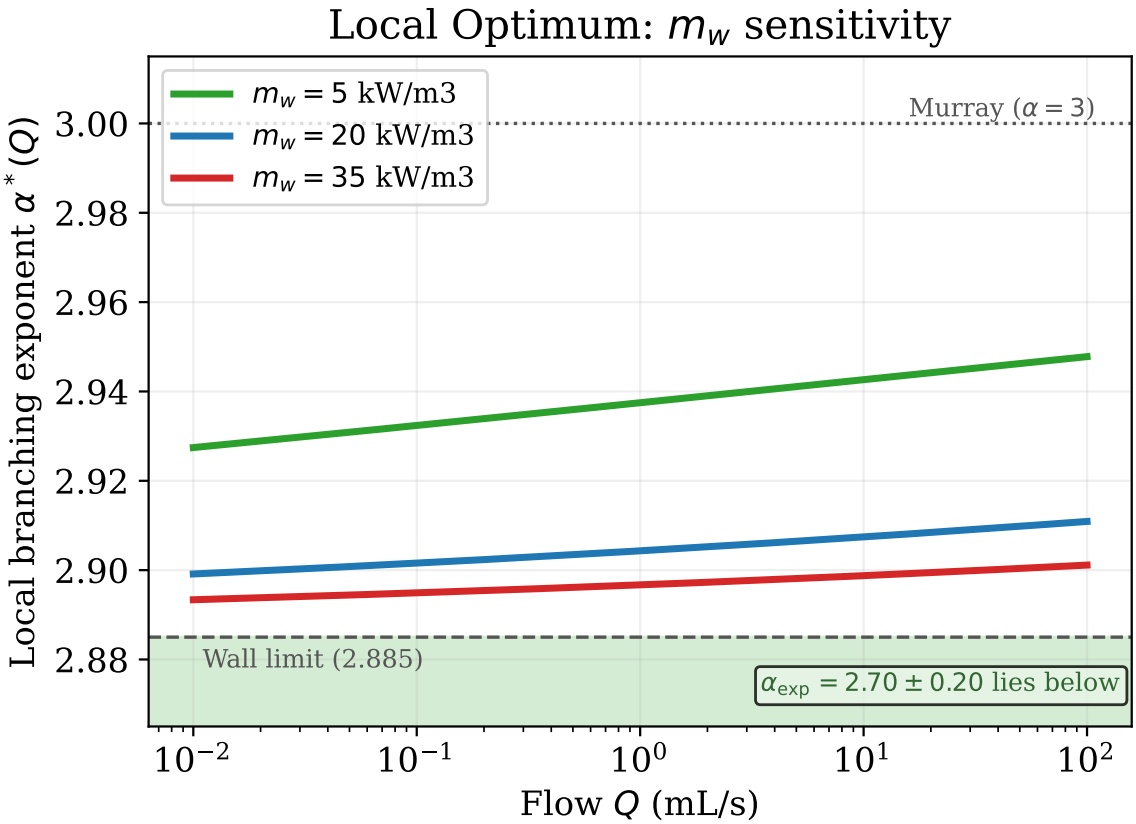

Figure 1: Scale-dependence of the local branching exponent $\alpha^*(Q)$. As flow Q increases, the exponent monotonically approaches the Murray limit ($\alpha = 3$) from below. Thicker walls (higher metabolic wall cost m_w) shift the entire curve toward the wall-dominated lower bound $(5+p)/2 \approx 2.885$. The shaded region marks the empirical morphometric range for porcine coronary arteries [2].

Table 2: Predicted α^* for porcine coronary arteries. No fitting; all parameters from independent literature.

m_w (kW/m^3)	r^* (mm)	α^*	Agreement with α_{exp}
5	1.200	2.938	within 1.2σ
20	1.019	2.905	within 1.0σ
35	0.937	2.897	within 1.0σ
Murray limit ($m_w \to 0$)		3.000000	1.5σ above mean
Experimental [2]		2.7 ± 0.2	–

σ computed from experimental uncertainty $\alpha_{\text{exp}} = 2.70 \pm 0.20$ [2]. All three-term predictions lie within 1–1.2σ of the measured mean, representing a measurable reduction relative to the Murray limit (1.5σ).

Table 3: Cross-network validation. Measured wall scaling exponent p and the associated single-term limit $\alpha = (5+p)/2$. When boundary thickness is governed exactly by vessel radius (e.g. Barlow's stress formula for internal pressure: $h \propto r$), $p = 1$ and the bounds collapse, recovering Murray's theoretical $\alpha = 3$ perfectly.

Network	p	α Limit	α_{exp}
Human pulmonary artery [14]	0.60	2.800	2.7–2.8
Porcine coronary artery [2, 6]	0.77	2.885	2.7±0.2
Plant xylem [15]	1.00	3.000 (Murray)	~ 3
Engineered pipes (pressure)	1.00	3.000 (Murray)	~ 3

The wall-thickness law also allows robust comparisons across highly disparate physical systems (Table 3). Human pulmonary arterial trees exhibit an independent morphometric exponent of $\alpha \approx 2.7$–2.8 [14]. Their characteristically thinner walls ($p \approx 0.60$) naturally push the theoretical limit lower than the systemic circulation, matching the direction of the empirical shift without requiring parameter fitting. Conversely, for networks designed statically to withstand internal pressure (engineered pipes or certain plant xylem [15]), mechanics dictate $h \propto r^1$. Setting $p = 1$ forces the bounds to collapse: $(5+1)/2 = 3 \leq \alpha^* \leq 3$. The optimization forces Murray's law to be recovered exactly.

7 Testable Predictions

1. *Scale gradient.* Since $\lambda \propto r_0^{p-1}$ is a strictly decreasing function of r_0 (and hence of Q) for $p < 1$, and ρ^* is strictly decreasing in λ (by implicit differentiation of (6)), $\alpha^*(Q) = \log 2/\log(1/\rho^*)$ is strictly increasing in Q. Thus $\alpha^*(Q)$ monotonically approaches 3 from below as Q increases (Corollary 7). Distal bifurcations (small Q) should have slightly smaller α than proximal bifurcations. Predicted range: $\Delta\alpha^* < 0.015$ across four decades.

2. *Wall-cost sensitivity.* Thicker-walled vessels (higher h/r, i.e. larger C/B) should have smaller α^*, approaching the wall-cost limit $(5+p)/2$.

3. *Physical grounding of γ.* Our Theorem 3 gives $\alpha = (4+\gamma)/2$ where $\gamma = 1+p$ is the wall-cost exponent. The same formula was obtained by Bennett [5] as $\alpha = (m+4)/2$ with m as a free pricing parameter. Our framework predicts $m = 1+p$ for any network whose maintenance cost is dominated by a wall-like sheath. Networks with known p (from histology or materials data) can test this prediction without morphometric fitting. We emphasize that this physical identification $\gamma = 1+p$, and hence the Bennett formula $\alpha = (4+\gamma)/2$, holds rigorously only in the two-term limit where either $B = 0$ or $C = 0$. In the physically complete three-term case ($B, C > 0$), Corollary 6 guarantees that no universal exponent exists and the Bennett single-index formula does not apply.

8 Relation to Prior Work

Murray [1]: two-term cost, $\alpha = 3$ (exact). Our Corollary 6 shows this is the *only* instantiation of the cost-function family (3) with a universal exponent.

Bennett [5]: changes the *objective* from energy to entropy-per-information-cost (EPIC), obtaining $\alpha = (m+4)/2$. We change the *structure* of the cost function, obtaining the same formula with $\gamma = 1+p$ physically grounded. In a parallel work [16], Bennett derives the identical index via a "chi index" formalism, establishing universality *specifically* within the two-term constraint.

Our Theorem 4 and Corollary 6 go exactly beyond this, breaking the constraint: we prove that incorporating the necessary third biological term (the $h(r) \propto r^p$ sublinear conduit tissue) shatters universal exponents altogether—a non-universal regime inexpressible within solely two-term structures.

Liu & Kassab [17] and Kim & Wagenseil [18] previously incorporated metabolic wall costs into branching models, but explicitly concluded that the standard scaling relations ($\alpha = 3$) remained largely invariant. This discrepancy underscores our central thesis: it is specifically the *sub-linear* allometric scaling of the wall ($p < 1$) that breaks the cubic law. Without it, standard three-term energy models natively recover Murray's limits.

Taylor *et al.* [19] report optimal Murray exponents of $P = 2.15$ (for volumetric flow) and $P = 2.38$ (for microvascular resistance) in human epicardial coronary arteries, using a CFD model constrained by pressure-wire measurements. We note that these values lie below our theoretical lower bound $(5 + p)/2 \approx 2.885$. This is not a contradiction: Taylor *et al.*'s exponents are hemodynamic optimality exponents—the best-fit P in the relationship $Q \propto D^P$ that reproduces *in-vivo* flow distributions in a clinical population with coronary disease—rather than morphometric diameter-scaling exponents in the sense of Eq. (1). The two quantities coincide only if the tree were exactly Murray-optimal; in diseased or developmentally constrained vascular beds the hemodynamic exponent reflects additional physiological constraints absent from the static cost function. The morphometric exponent most directly comparable to our framework is that of Kassab *et al.* [2], $\alpha_{\text{exp}} = 2.7 \pm 0.2$, measured from perfusion-fixed casts of healthy porcine coronary trees, which lies within our predicted range. The sub-2.885 values reported by Taylor *et al.* suggest that pulsatile wave costs introduce an additional architectural constraint not captured by the static cost function analyzed here.

Smink *et al.* [20]: extend Murray's two-term optimization to turbulent flows, non-Newtonian fluids, and rough-wall channels, deriving the exponent x in $Q \propto r^x$ across the full Reynolds-number parameter space. Their framework covers the fluid-rheology dimension of the optimization problem, while ours covers the biological wall-tissue dimension. Neither framework addresses the signal-transport dimension (such as pulsatile wave reflection in arteries or electrical signal attenuation in neurons), which we propose as the final axis required for a complete universal classification.

West, Brown & Enquist [21]: minimize total network resistance over a self-similar infinite tree, obtaining Kleiber's 3/4 law. Our optimization is at the single-vessel level; the network structure enters only through Murray's branching condition (Lemma 2).

Rall [22]: the branching exponent $\alpha = 3/2$ for electrotonic signal propagation in neurons is derived from impedance-matching in cable theory ($Z \propto r^{-3/2}$), not from cost minimization. Corollary 6 therefore does not contradict Rall: the two universality results arise from physically distinct mechanisms (energy minimization vs. impedance matching) and cover non-overlapping parameter ranges. However, this mathematical structure strongly suggests that a generalized Lagrangian unifying metabolic maintenance cost with signal attenuation impedance could natively bridge vascular and neural morphologies.

9 Conclusion

We have shown that the three-term cost function (2) leads to several rigorous results unavailable in the existing Murray–Bennett framework: unique optimal radius (Theorem 1), equivalence of local optimization with global tree law (Lemma 2), exact single-term classification $\alpha = (4 + \gamma)/2$ with physical grounding of γ (Theorem 3), strict non-universality of the three-term branching exponent extending to asymmetric flows (Theorem 4, Corollary 5), breaking of structural degeneracy to bound the optimal topology to small branching integers, excluding star-like

multifurcations (Theorem 10), and tight theoretical bounds on the optimal symmetric bifurcation angle, independently confirmed by three-dimensional morphometric data ($74.9^\circ < 2\theta^* < 80.2^\circ$), parameterizing the deviation from Murray's geometry (Theorem 9). The central prediction $\alpha^* \in [2.90, 2.94]$ from independently measured parameters measurably reduces the gap between Murray's cubic law and cardiovascular data.

Static vs. pulsatile contributions. The purely static wall-tissue mechanism predicts $\alpha^* \approx 2.90$ for porcine coronary arteries, already indicating a structural symmetry-breaking from Murray's law and narrowing—but not closing—the gap with cardiovascular morphometric data. However, arterial networks are not merely static pipes; they are dynamic conduits subject to pulsatile wave reflection, which independently favors impedance matching at bifurcations ($\alpha \to 2$). The rigorous non-universality proven here establishes that the final physiological exponent cannot be strictly cubic, but must instead emerge from a competitive balance between these disparate thermodynamic demands. Bridging the residual gap between our exact static derivation (~ 2.90) and the empirical clinical reality ($\alpha \approx 2.5$–2.7) therefore mathematically necessitates integrating both viscous tissue dissipation and wave-reflection energetics into a single, unified network-level Lagrangian. Given the strict bounds on single-mechanism scaling established in this work, formulating such a unified network-level variational principle represents the necessary definitive step to predict both fluidic cardiovascular architecture and electrotonic neural morphology.

Data and Code Availability. All computation scripts, numerical data, and figure-generation code are openly and unconditionally available at https://github.com/rikymarche-ctrl/vascular-networks-theory under the MIT Licence. No access request is required.

References

[1] C. D. Murray. The physiological principle of minimum work. *Proc. Natl. Acad. Sci. U.S.A.*, 12:207–214, 1926. https://doi.org/10.1073/pnas.12.3.207.

[2] G. S. Kassab, C. A. Rider, N. J. Tang, and Y. C. Fung. Morphometry of pig coronary arterial trees. *Am. J. Physiol.*, 265:H350–H365, 1993. https://doi.org/10.1152/ajpheart.1993.265.1.H350.

[3] M. Zamir and W. C. Bigelow. Cost of departure from optimality in arterial branching. *J. Theor. Biol.*, 197:517–523, 1999. https://doi.org/10.1016/S0022-5193(84)80089-2.

[4] W. W. Nichols, M. F. O'Rourke, and C. Vlachopoulos. *McDonald's Blood Flow in Arteries*. Hodder Arnold, London, 6th edition, 2011. https://doi.org/10.1201/b13568.

[5] J. Bennett. Murray's law as an entropy-per-information-cost extremum. *arXiv:2511.04022*, 2024. preprint, https://arxiv.org/abs/2511.04022.

[6] J. A. G. Rhodin. The ultrastructure of mammalian arterioles and precapillary sphincters. *J. Ultrastruct. Res.*, 18:181–223, 1967. https://doi.org/10.1016/S0022-5320(67)80239-9.

[7] H. Wolinsky and S. Glagov. A lamellar unit of aortic medial structure and function in mammals. *Circ. Res.*, 20:99–111, 1967. https://doi.org/10.1161/01.res.20.1.99.

[8] C. G. Caro, T. J. Pedley, R. C. Schroter, and W. A. Seed. *The Mechanics of the Circulation*. Oxford University Press, Oxford, 1978. https://www.worldcat.org/isbn/0192633236.

[9] L. A. Taber. An optimization principle for vascular radius including the effects of smooth muscle tone. *Biophys. J.*, 74:109–114, 1998. https://doi.org/10.1016/S0006-3495(98)77772-0.

[10] R. J. Paul. Chemical energetics of vascular smooth muscle. In *Handbook of Physiology*, volume 2, pages 201–235. 1980. https://doi.org/10.1002/cphy.cp020209.

[11] M. Zamir. Nonsymmetrical bifurcations in arterial branching. *J. Gen. Physiol.*, 72:837–845, 1978. https://doi.org/10.1085/jgp.72.6.837.

[12] M. Zamir and J. A. Medeiros. Arterial branching in man and monkey. *J. Gen. Physiol.*, 79(3):353–360, 1982. https://doi.org/10.1085/jgp.79.3.353.

[13] B. Kaimovitz, Y. Lanir, and G. S. Kassab. Large-scale 3-d geometric reconstruction of the porcine coronary arterial vasculature based on detailed anatomical data. *Ann. Biomed. Eng.*, 33(11):1517–1535, 2005. https://doi.org/10.1007/s10439-005-7544-3.

[14] W. Huang, R. T. Yen, M. McLaurine, and G. Bledsoe. Morphometry of the human pulmonary vasculature. *J. Appl. Physiol.*, 81:2123–2133, 1996. https://doi.org/10.1152/jappl.1996.81.5.2123.

[15] U. G. Hacke, J. S. Sperry, W. T. Pockman, S. D. Davis, and K. A. McCulloh. Trends in wood density and structure are linked to prevention of xylem cavitation by negative pressure. *Oecologia*, 126:457–461, 2001. https://doi.org/10.1007/s004420100628.

[16] J. Bennett. A single-index theory of optimal branching. *arXiv:2511.19915*, 2024. preprint, https://arxiv.org/abs/2511.19915.

[17] Y. Liu and G. S. Kassab. Vascular metabolic dissipation in murray's law. *Am. J. Physiol. Heart Circ. Physiol.*, 292(3):H1336–H1339, 2007. https://doi.org/10.1152/ajpheart.00906.2006.

[18] J. Kim and J. E. Wagenseil. Bio-chemo-mechanical models of vascular mechanics. *Ann. Biomed. Eng.*, 43(7):1477–1487, 2015. https://doi.org/10.1007/s10439-014-1201-7.

[19] D. J. Taylor, J. Feher, I. Halliday, D. R. Hose, R. Gosling, L. Aubiniere-Robb, et al. Refining our understanding of the flow through coronary artery branches; revisiting murray's law in human epicardial coronary arteries. *Front. Physiol.*, 13:871912, 2022. https://doi.org/10.3389/fphys.2022.871912.

[20] J. S. Smink, R. Hagmeijer, C. H. Venner, and C. W. Visser. Optimising branched fluidic networks: a unifying approach including laminar and turbulent flows, rough walls and non-newtonian fluids. *J. Fluid Mech.*, 1011:A42, 2025. https://doi.org/10.1017/jfm.2025.393.

[21] G. B. West, J. H. Brown, and B. J. Enquist. A general model for the origin of allometric scaling laws in biology. *Science*, 276:122–126, 1997. https://doi.org/10.1126/science.276.5309.122.

[22] W. Rall. Branching dendritic trees and motoneuron membrane resistivity. *Exp. Neurol.*, 1:491–527, 1959. https://doi.org/10.1016/0014-4886(59)90046-9.